

Resolve smoothed complexity of pure Nash equilibria in unrestricted congestion games

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Website #28 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

We are given an unweighted congestion game with players $N = [n]$, resources R , strategy sets $\Sigma_i \subseteq 2^R$, and nonnegative latency functions $\kappa_r : [n] \rightarrow \mathbb{R}_{\geq 0}$. For a profile $\sigma \in \Sigma := \prod_i \Sigma_i$, the load on r is $\ell_r(\sigma)$, player i 's cost is

$$C_i(\sigma) = \sum_{r \in \sigma_i} \kappa_r(\ell_r(\sigma)),$$

and Rosenthal's potential is

$$\Phi(\sigma) = \sum_{r \in R} \sum_{\ell=1}^{\ell_r(\sigma)} \kappa_r(\ell).$$

Every better-response move strictly decreases Φ , hence better-response dynamics (BRD) always terminate at a pure Nash equilibrium (PNE).

The question asks for the *smoothed* complexity of exact BRD under three perturbation models:

1. **General latencies:** each value $\kappa_r(\ell) \in [0, 1]$ is independently drawn from a continuous distribution with density at most ϕ .
2. **Polynomial latencies:** $\kappa_r(\ell) = \sum_{j=0}^{d_r} \alpha_{r,j} \ell^j$, with each coefficient $\alpha_{r,j} \in [0, 1]$ independently drawn with density at most ϕ .
3. **Step-function latencies:** $\kappa_r(\ell) = \sum_{j: b_{r,j} \leq \ell} \alpha_{r,j}$, where the breakpoints $b_{r,j}$ are adversarially fixed and the jump sizes $\alpha_{r,j} \in [0, 1]$ are independently drawn with density at most ϕ .

The hardest reading is:

For unrestricted explicit congestion games (and, by extension, unrestricted network congestion games under shortest-path dynamics), is the expected number of exact better-response steps polynomial in the input size and ϕ ? If not, what structural conditions precisely separate polynomial from superpolynomial smoothed running time?

Its negation would be: there exists a family of smoothed congestion games, under one of the three perturbation models, for which the expected number of exact better-response steps is superpolynomial in the input size and ϕ .

Alternative readings considered

Two possible ambiguities matter.

1. *Exact vs. approximate equilibrium.* I adopt the harder exact-PNE reading. I only mention approximate PNE as related work.

2. *Explicit vs. network representation.* I first treat explicit congestion games, then derive a compact-network corollary for shortest-path dynamics.

I do *not* weaken the problem to restrained/compact games as an interpretation choice; rather, I prove a rigorous partial theorem showing that a concrete structural parameter yields polynomial or quasipolynomial smoothed complexity, and I explain the remaining gap to the unrestricted exact case.

Related literature found

- **Giannakopoulos–Grosz–Melissourgos (2025), *PARTIAL*.** This paper proves exact polynomial smoothed time for B -restrained explicit congestion games and for (A, B) -compact network congestion games, under the same three perturbation models; it also shows these restricted classes remain PLS-hard, hence the restriction is nontrivial. I use it as orientation, not as a step in the proof below.
- **Giannakopoulos (EC 2024), *PARTIAL*.** This paper proves a smoothed FPTAS for $(1 + \varepsilon)$ -approximate PNE in general/step/polynomial congestion games: any $(1 + \varepsilon)$ -improving dynamics converges in expected time polynomial in ϕ , $1/\varepsilon$, and the game description. This shows that the exactness requirement is the main unresolved feature in the unrestricted setting.
- **Fabrikant–Papadimitriou–Talwar (2004), *BACKGROUND*.** General congestion games are PLS-complete in the worst case.
- **Ackermann–Röglin–Vöcking (2008), *BACKGROUND/PARTIAL*.** They show exponentially long best-response sequences for non-matroid strategy spaces and PLS-completeness for network congestion games even with linear latencies. I inspected these constructions as candidate smoothed lower bounds.
- **Chen–Guo–Vlatakis–Gkaragkounis–Yannakakis–Zhang (2019/2020), *TOOL that did not apply*.** Their rank-based smoothed analysis for FLIP / local Max-Cut / binary Max-CSP looked promising, but congestion-game potentials are not binary pairwise objectives; I could not transplant that machinery directly.

No primary source I found claimed a polynomial smoothed upper bound for *unrestricted exact* better-response dynamics in the three congestion-game models above. The exact positive results I found all impose restraint/compactness hypotheses, whereas the unrestricted general case is solved only for approximate equilibria.

Proof sketch

The proof below is independent of the cited papers.

Fix an explicit congestion game. For every ordered unilateral deviation (σ_i, τ_i) and every possible choice of the relevant loads on the resources in $\sigma_i \Delta \tau_i$, I define a *formal deviation gain*. It is a linear form in the perturbed numerical parameters. Every actual better-response gain is one of these formal gains.

The key quantitative parameter is therefore

$$Q(\Gamma) := \sum_{i=1}^n \sum_{\substack{\sigma, \tau \in \Sigma_i \\ \sigma \neq \tau}} n^{|\sigma \Delta \tau|},$$

because an ordered pair (σ, τ) generates at most $n^{|\sigma\Delta\tau|}$ formal gains.

Now fix one nonzero formal gain G . Since the perturbations are independent and continuous with density at most ϕ , conditioning on all random parameters except one with nonzero coefficient shows

$$\Pr(|G| \leq \varepsilon) \leq 2\phi\varepsilon.$$

A union bound over all formal gains gives a tail bound for the minimum positive improvement

$$\delta := \min\{|G| : G \neq 0 \text{ is a formal gain}\}.$$

If δ is not tiny, then every better-response step drops Rosenthal's potential by at least δ , so the number of steps is at most $\text{range}(\Phi)/\delta$.

A direct expectation of $1/\delta$ would diverge logarithmically. The fix is to combine the potential-drop bound with the trivial bound $T \leq |\Sigma| - 1$, since a better-response path never revisits a profile. This truncation converts the linear tail for δ into an $O(\log |\Sigma|)$ factor. The result is

$$\mathbb{E}[T] \leq O(\phi \cdot \text{range}(\Phi) \cdot Q(\Gamma) \cdot \log |\Sigma|).$$

This immediately yields polynomial smoothed time whenever $Q(\Gamma)$ is polynomially bounded, and quasipolynomial smoothed time whenever $Q(\Gamma)$ is quasipolynomially bounded. In particular, B -restrained games satisfy $Q(\Gamma) \leq nk^2n^B$, where $k = \max_i |\Sigma_i|$, giving polynomial bounds for constant B . A similar stage decomposition yields a compact-network corollary.

Main result

Theorem 1 (Explicit games: a quantitative sufficient condition). *Let Γ be an explicit n -player congestion game with resource set R , strategy sets $(\Sigma_i)_{i=1}^n$, and*

$$S := |\Sigma| = \prod_{i=1}^n |\Sigma_i|, \quad Q(\Gamma) := \sum_{i=1}^n \sum_{\substack{\sigma, \tau \in \Sigma_i \\ \sigma \neq \tau}} n^{|\sigma\Delta\tau|}.$$

Let T be the number of better-response moves, under an arbitrary pivoting rule and from an arbitrary initial profile, until a PNE is reached.

Then the following bounds hold.

1. General latencies.

$$\mathbb{E}[T] \leq 1 + 2\phi mn Q(\Gamma) \log S.$$

2. Polynomial latencies of maximum degree d .

$$\mathbb{E}[T] \leq 1 + 2\phi m(d+1)n^{d+1} Q(\Gamma) \log S.$$

3. Step-function latencies with at most d breakpoints per resource.

$$\mathbb{E}[T] \leq 1 + 2\phi m d n Q(\Gamma) \log S.$$

Corollary 1 (Restrained explicit games). *Suppose Γ is B -restrained, i.e.*

$$\max_i \max_{\sigma, \tau \in \Sigma_i} |\sigma\Delta\tau| \leq B,$$

and write $k := \max_i |\Sigma_i|$. Then

$$Q(\Gamma) \leq nk^2 n^B,$$

so

$$\mathbb{E}[T] \leq \begin{cases} 1 + 2\phi m k^2 n^{B+2} \log S, & \text{general,} \\ 1 + 2\phi m(d+1) k^2 n^{B+d+2} \log S, & \text{polynomial,} \\ 1 + 2\phi m d k^2 n^{B+2} \log S, & \text{step-function.} \end{cases}$$

Hence, in explicit representation:

- for the **general** and **step-function** models, constant B implies polynomial smoothed time;
- for the **polynomial** model, constant B and constant degree d imply polynomial smoothed time;
- if $B = O(\log N)$ (and $d = O(\log N)$ in the polynomial model), the bound is quasipolynomial in the input size N and ϕ .

Corollary 2 (Compact network games). *Let Γ be an (A, B) -compact network congestion game, and run shortest-path dynamics from an arbitrary initial profile. Then the expected number T_{sp} of iterations satisfies*

$$\mathbb{E}[T_{\text{sp}}] = \begin{cases} O(\phi m A^2 n^{2B+4} \log A), & \text{general,} \\ O(\phi m(d+1) A^2 n^{2B+d+4} \log A), & \text{polynomial,} \\ O(\phi m d A^2 n^{2B+4} \log A), & \text{step-function.} \end{cases}$$

In particular, if A is polynomially bounded and $B = O(1)$, then shortest-path dynamics has polynomial smoothed complexity in the general and step-function models, and also in the polynomial model when $d = O(1)$.

Proof

Lemma 1 (Exact potential identity). *Let $\sigma = (\sigma_i, \sigma_{-i})$ and let $\tau = (\tau_i, \sigma_{-i})$ differ only in player i 's strategy. Then*

$$\Phi(\sigma) - \Phi(\tau) = \sum_{r \in \sigma_i \setminus \tau_i} \kappa_r(\ell_r(\sigma)) - \sum_{r \in \tau_i \setminus \sigma_i} \kappa_r(\ell_r(\sigma) + 1) = C_i(\sigma) - C_i(\tau).$$

Proof. Fix a resource r .

If $r \notin \sigma_i \Delta \tau_i$, its load is the same in σ and τ , hence its contribution to $\Phi(\sigma) - \Phi(\tau)$ is 0.

If $r \in \sigma_i \setminus \tau_i$, then the load drops from $\ell_r(\sigma)$ to $\ell_r(\sigma) - 1$. Therefore

$$\sum_{\ell=1}^{\ell_r(\sigma)} \kappa_r(\ell) - \sum_{\ell=1}^{\ell_r(\sigma)-1} \kappa_r(\ell) = \kappa_r(\ell_r(\sigma)).$$

If $r \in \tau_i \setminus \sigma_i$, then the load rises from $\ell_r(\sigma)$ to $\ell_r(\sigma) + 1$. Therefore

$$\sum_{\ell=1}^{\ell_r(\sigma)} \kappa_r(\ell) - \sum_{\ell=1}^{\ell_r(\sigma)+1} \kappa_r(\ell) = -\kappa_r(\ell_r(\sigma) + 1).$$

Summing over all resources gives the first displayed identity.

For the second identity, note that

$$C_i(\sigma) = \sum_{r \in \sigma_i \cap \tau_i} \kappa_r(\ell_r(\sigma)) + \sum_{r \in \sigma_i \setminus \tau_i} \kappa_r(\ell_r(\sigma)),$$

whereas

$$C_i(\tau) = \sum_{r \in \sigma_i \cap \tau_i} \kappa_r(\ell_r(\sigma)) + \sum_{r \in \tau_i \setminus \sigma_i} \kappa_r(\ell_r(\sigma) + 1).$$

Subtracting yields the claim. $\square$

Lemma 2 (Anti-concentration for one linear form). *Let $X_1, \dots, X_q$ be independent random variables, each supported on $[0, 1]$ with density bounded by ϕ . Let*

$$G = b + \sum_{t=1}^q c_t X_t$$

with at least one $c_t \neq 0$. Then for every $\varepsilon > 0$,

$$\Pr(|G| \leq \varepsilon) \leq 2\phi\varepsilon.$$

Proof. Choose an index u with $c_u \neq 0$. Condition on all variables X_t with $t \neq u$. Then G becomes

$$G = \beta + c_u X_u$$

for some deterministic β . Hence the event $|G| \leq \varepsilon$ forces X_u to lie in an interval of length $2\varepsilon/|c_u|$. Since X_u has density at most ϕ ,

$$\Pr(|G| \leq \varepsilon \mid X_t, t \neq u) \leq \phi \cdot \frac{2\varepsilon}{|c_u|} \leq 2\phi\varepsilon.$$

Taking expectation over the conditioned variables yields the same bound unconditionally. $\square$

Lemma 3 (Formal deviation gains). *For each model, define the family of formal deviation gains as follows.*

1. **General model.** For $i \in [n]$, $\sigma, \tau \in \Sigma_i$ with $\sigma \neq \tau$, and $\xi : \sigma \Delta \tau \rightarrow [n]$,

$$G_{i,\sigma,\tau,\xi}^{\text{gen}} := \sum_{r \in \sigma \setminus \tau} \kappa_r(\xi_r) - \sum_{r \in \tau \setminus \sigma} \kappa_r(\xi_r).$$

2. **Polynomial model.** For the same data,

$$G_{i,\sigma,\tau,\xi}^{\text{poly}} := \sum_{r \in \sigma \setminus \tau} \sum_{j=0}^{d_r} \alpha_{r,j} \xi_r^j - \sum_{r \in \tau \setminus \sigma} \sum_{j=0}^{d_r} \alpha_{r,j} \xi_r^j.$$

3. **Step-function model.** Writing $\kappa_r(\ell) = \sum_{j: b_{r,j} \leq \ell} \alpha_{r,j}$, define

$$G_{i,\sigma,\tau,\xi}^{\text{step}} := \sum_{r \in \sigma \setminus \tau} \sum_{j: b_{r,j} \leq \xi_r} \alpha_{r,j} - \sum_{r \in \tau \setminus \sigma} \sum_{j: b_{r,j} \leq \xi_r} \alpha_{r,j}.$$

Then every actual unilateral deviation gain

$$C_i(\boldsymbol{\sigma}) - C_i(\tau_i, \boldsymbol{\sigma}_{-i}) = \Phi(\boldsymbol{\sigma}) - \Phi(\tau_i, \boldsymbol{\sigma}_{-i})$$

is equal to one member of the corresponding formal family.

Moreover, the number of formal gains is at most

$$Q(\Gamma) = \sum_{i=1}^n \sum_{\substack{\sigma, \tau \in \Sigma_i \\ \sigma \neq \tau}} n^{|\sigma \Delta \tau|}.$$

Proof. Fix an actual deviation from σ_i to τ_i at profile $\boldsymbol{\sigma}$. For $r \in \sigma_i \setminus \tau_i$, let $\xi_r := \ell_r(\boldsymbol{\sigma}) \in [n]$. For $r \in \tau_i \setminus \sigma_i$, let $\xi_r := \ell_r(\boldsymbol{\sigma}) + 1 \in [n]$.

By the previous lemma,

$$\Phi(\boldsymbol{\sigma}) - \Phi(\tau_i, \boldsymbol{\sigma}_{-i}) = \sum_{r \in \sigma_i \setminus \tau_i} \kappa_r(\xi_r) - \sum_{r \in \tau_i \setminus \sigma_i} \kappa_r(\xi_r),$$

which is exactly $G_{i, \sigma_i, \tau_i, \xi}^{\text{gen}}$ in the general model. Expanding $\kappa_r(\xi_r)$ in terms of the perturbed coefficients gives the polynomial and step-function formulas.

For counting: fixing i, σ, τ , the map ξ has domain $\sigma \Delta \tau$, so there are at most $n^{|\sigma \Delta \tau|}$ choices. Summing over all ordered pairs (σ, τ) with $\sigma \neq \tau$ gives at most $Q(\Gamma)$ formal gains. $\square$

Lemma 4 (Tail bound for the minimum positive gain). *Fix one of the three perturbation models. Let $\mathcal{G}$ be the corresponding finite family of formal gains, and define*

$$\delta := \min\{|G| : G \in \mathcal{G}, G \neq 0\}.$$

Then for every $\varepsilon > 0$,

$$\Pr(\delta \leq \varepsilon) \leq 2\phi Q(\Gamma) \varepsilon.$$

Proof. Every nonzero $G \in \mathcal{G}$ is a linear form in independent perturbed parameters.

- In the *general* model, the coefficients are in $\{-1, 0, 1\}$, and any changed resource contributes one variable $\kappa_r(\xi_r)$ with coefficient ± 1 .
- In the *polynomial* model, for any changed resource r , the coefficient of $\alpha_{r,0}$ is ± 1 , so any nontrivial formal gain has a random variable with nonzero coefficient.
- In the *step-function* model, a formal gain can be zero as a formal expression if every changed resource is below all active breakpoints; those formal gains are excluded by the definition of δ . Any nonzero formal gain contains some jump variable $\alpha_{r,j}$ with coefficient ± 1 .

Hence each nonzero $G \in \mathcal{G}$ satisfies

$$\Pr(|G| \leq \varepsilon) \leq 2\phi \varepsilon$$

by the anti-concentration lemma. A union bound over at most $Q(\Gamma)$ formal gains gives

$$\Pr(\delta \leq \varepsilon) \leq \sum_{\substack{G \in \mathcal{G} \\ G \neq 0}} \Pr(|G| \leq \varepsilon) \leq 2\phi Q(\Gamma) \varepsilon.$$

$\square$

Lemma 5 (Truncated reciprocal bound). *Let $R > 0$, $S \geq 1$, and $\delta > 0$ be random. Suppose*

$$T \leq \min\{S, R/\delta\} \quad \text{almost surely}$$

and

$$\Pr(\delta \leq \varepsilon) \leq a\varepsilon \quad \text{for all } \varepsilon > 0.$$

Then

$$\mathbb{E}[T] \leq 1 + aR \log S.$$

Proof. Using $\mathbb{E}[T] = \int_0^\infty \Pr(T \geq t) dt$ and $T \leq S$,

$$\mathbb{E}[T] = \int_0^S \Pr(T \geq t) dt \leq 1 + \int_1^S \Pr(T \geq t) dt.$$

If $T \geq t$ for some $t \geq 1$, then $\min\{S, R/\delta\} \geq t$, hence $R/\delta \geq t$, i.e.

$$\delta \leq \frac{R}{t}.$$

Therefore

$$\mathbb{E}[T] \leq 1 + \int_1^S \Pr\left(\delta \leq \frac{R}{t}\right) dt \leq 1 + aR \int_1^S \frac{dt}{t} = 1 + aR \log S.$$

□

Proof of the main theorem. Fix one of the three perturbation models.

Let δ be the minimum positive absolute value among all corresponding formal gains. By the lemma on formal gains, every actual better-response move has gain equal to one of these formal gains. Hence every actual better-response step decreases Φ by at least δ .

Also, because Φ strictly decreases along better responses, no profile can repeat, so

$$T \leq S - 1 \leq S.$$

Thus it remains to bound the range of Φ .

General model. Since each $\kappa_r(\ell) \in [0, 1]$,

$$0 \leq \Phi(\sigma) \leq \sum_{r \in R} \sum_{\ell=1}^n 1 = mn.$$

Hence $T \leq \min\{S, mn/\delta\}$. Using the previous two lemmas with $a = 2\phi Q(\Gamma)$ and $R = mn$,

$$\mathbb{E}[T] \leq 1 + 2\phi mn Q(\Gamma) \log S.$$

Polynomial model. For every resource r and load $\ell \in [n]$,

$$\kappa_r(\ell) = \sum_{j=0}^{d_r} \alpha_{r,j} \ell^j \leq \sum_{j=0}^d \ell^j \leq (d+1)n^d.$$

Therefore

$$0 \leq \Phi(\sigma) \leq \sum_{r \in R} \sum_{\ell=1}^n (d+1)n^d = m(d+1)n^{d+1}.$$

Hence $T \leq \min\{S, m(d+1)n^{d+1}/\delta\}$, and so

$$\mathbb{E}[T] \leq 1 + 2\phi m(d+1)n^{d+1} Q(\Gamma) \log S.$$

Step-function model. For every resource r and load ℓ , at most $d_r \leq d$ jumps contribute, each of size at most 1. Thus

$$\kappa_r(\ell) \leq d,$$

and therefore

$$0 \leq \Phi(\sigma) \leq \sum_{r \in R} \sum_{\ell=1}^n d = mdn.$$

Hence $T \leq \min\{S, mdn/\delta\}$, and

$$\mathbb{E}[T] \leq 1 + 2\phi mdn Q(\Gamma) \log S.$$

This proves the theorem. $\square$

Proof of the restrained explicit-game corollary. If Γ is B -restrained, then for every ordered pair (σ, τ) of strategies of the same player,

$$|\sigma \Delta \tau| \leq B.$$

Therefore

$$Q(\Gamma) = \sum_{i=1}^n \sum_{\substack{\sigma, \tau \in \Sigma_i \\ \sigma \neq \tau}} n^{|\sigma \Delta \tau|} \leq n^B \sum_{i=1}^n |\Sigma_i|(|\Sigma_i| - 1) \leq n^B \cdot nk^2.$$

Substituting this into the theorem yields the displayed bounds.

In explicit representation, n, m, k, B (and d , when relevant) are at most polynomial in the input length N . Hence constant B gives polynomial smoothed time in the general and step-function models, and constant B, d give polynomial smoothed time in the polynomial model. If $B = O(\log N)$, then $n^B = N^{O(\log N)}$, giving quasipolynomial time; likewise if $B + d = O(\log N)$ in the polynomial model. $\square$

Proof of the compact network corollary. Let the witness sets from (A, B) -compactness be $\Sigma_i^* \subseteq \Sigma_i$, so that every best response of player i lies in Σ_i^* , and every strategy in Σ_i^* has path length at most B .

Run shortest-path dynamics. Once a player best-responds for the first time, she moves into Σ_i^* , and every later best response of that player remains in Σ_i^* .

Relabel players in the order in which they first move. For $t = 1, \dots, n$, consider the time interval during which exactly the first t relabeled players have already moved at least once, but player $t+1$ has not yet moved (or the game has already terminated). During this interval:

- only players $1, \dots, t$ can move;
- each such player moves among strategies in Σ_i^* ;
- every ordered pair of strategies in Σ_i^* has symmetric difference at most $2B$, because each path has at most B edges.

Fixing the other players' current paths as background, the moves of players $1, \dots, t$ in this interval form a better-response sequence in an explicit congestion game on the same edge set, with strategy sets $\Sigma_1^*, \dots, \Sigma_t^*$. For this explicit subgame,

$$Q_t \leq \sum_{i=1}^t |\Sigma_i^*|^2 n^{2B} \leq t A^2 n^{2B}, \quad S_t \leq A^t.$$

Applying the main theorem to this t -player subgame gives

$$\mathbb{E}[T_t] \leq \begin{cases} 1 + 2\phi mn \cdot t A^2 n^{2B} \cdot t \log A, & \text{general,} \\ 1 + 2\phi m(d+1)n^{d+1} \cdot t A^2 n^{2B} \cdot t \log A, & \text{polynomial,} \\ 1 + 2\phi mdn \cdot t A^2 n^{2B} \cdot t \log A, & \text{step-function,} \end{cases}$$

where T_t denotes the number of steps the interval would need to reach equilibrium if no new player were introduced earlier.

The actual interval is no longer than T_t , and there are at most n first moves total. Hence

$$T_{\text{sp}} \leq n + \sum_{t=1}^n T_t.$$

Using $\sum_{t=1}^n t^2 = O(n^3)$, we obtain

$$\mathbb{E}[T_{\text{sp}}] = \begin{cases} O(\phi m A^2 n^{2B+4} \log A), & \text{general,} \\ O(\phi m(d+1) A^2 n^{2B+d+4} \log A), & \text{polynomial,} \\ O(\phi md A^2 n^{2B+4} \log A), & \text{step-function.} \end{cases}$$

This proves the corollary. □

Gap to the full statement

What remains unproved is the hardest reading: whether *every unrestricted exact* congestion game under the three smoothed perturbation models has $\mathbb{E}[T] \leq \text{poly}(N, \phi)$, where N is the input size.

The theorem above gives a rigorous sufficient condition: polynomial (or quasipolynomial) smoothed complexity follows whenever the formal-gain count $Q(\Gamma)$ is polynomially (or quasipolynomially) bounded. This covers restrained explicit games and compact network games, but it does not settle arbitrary unrestricted instances.

The literature I found likewise leaves the unrestricted exact case open: exact positive results are proved for B -restrained / (A, B) -compact classes, while unrestricted general instances are presently covered only for $(1 + \varepsilon)$ -approximate equilibria.

So the strongest fully rigorous conclusion I can currently justify is:

Exact BRD is polynomial in (N, ϕ) under concrete structural conditions (e.g. constant restraint / compactness, or more generally polynomially bounded $Q(\Gamma)$); the unrestricted exact smoothed complexity is not settled by the proof here, nor by any primary source I found.

Self red-team

Counterexample attempts.

1. *Could a step-function deviation have positive gain although every changed resource contributes zero random coefficients?* No. If every coefficient were zero, the gain would be identically zero, so the deviation could not be improving.
2. *Could the same random variable appear with opposite signs and cancel inside one formal gain?* Not in the explicit-game theorem: a resource belongs either to $\sigma \setminus \tau$ or to $\tau \setminus \sigma$, never both, so in the general/step models each changed resource contributes at most once. In the polynomial model, the constant-term coefficient ± 1 for a changed resource prevents complete cancellation of that resource's contribution.
3. *Does $\mathbb{E}[R/\delta]$ diverge?* Yes; that is exactly why the truncation by $T \leq S$ is essential. I explicitly checked that the proof uses $\min\{S, R/\delta\}$, not R/δ alone.

Three plausible referee objections.

1. *“You overcount infeasible load assignments ξ .”* Correct, but harmless: overcounting only enlarges $Q(\Gamma)$, hence only weakens the upper bound.
2. *“In the network corollary, background loads shift the latency functions.”* Also correct, but the proof never requires the shifted coefficients to be independent. It applies the explicit-game theorem to actual move gains viewed as linear forms in the *original* independent perturbed parameters.
3. *“Polynomial latencies might have degree d growing with the input.”* Yes. The theorem is still correct, but polynomial smoothed time then requires n^d to be polynomially bounded by the input size; in particular, constant d suffices.

Boundary checks.

- If every player has one strategy, then $S = 1$, $T = 0$, and the bound is vacuous but valid.
- If $B = 0$, then $Q(\Gamma) = 0$, so no deviation exists; again $T = 0$.
- If $n = 1$, every better-response path has length at most $|\Sigma_1| - 1$, consistent with the theorem.
- For step-functions with $d = 0$, the model has zero latency everywhere, so $T = 0$; the bound still holds.

Ideas tried

1. Direct minimum-improvement argument for unrestricted exact BRD.

Idea: enumerate all possible unilateral gains, anti-concentrate each, union bound, and divide the potential range by the minimum positive gain.

What worked: this yielded the theorem above, with the structural parameter $Q(\Gamma)$.

Obstruction: for unrestricted games, $Q(\Gamma)$ can be exponentially large because one ordered strategy pair (σ, τ) contributes $n^{|\sigma \Delta \tau|}$. This blocks a general polynomial bound by this route.

2. **Import rank-based long-sequence methods from smoothed FLIP / Max-CSP.**

Idea: mimic the rank arguments used for local Max-Cut / binary Max-CSP, where long improvement paths are unlikely because the corresponding matrix has large rank.

Obstruction: congestion-game potentials are not binary pairwise objectives; the dependence on the strategy profile is via resource loads and induces higher-order interactions among players. I could not identify a comparable rank object for exact congestion BRD.

3. **Robustify known worst-case exponential lower bounds into smoothed lower bounds.**

Idea: start from the binary-counter constructions of Ackermann–Röglin–Vöcking and place each perturbed parameter in a tiny interval around the deterministic one.

Obstruction: the standard construction uses exponentially separated scales $c_i = \alpha^{20i}$. After normalizing parameters into $[0, 1]$, the smallest relevant magnitudes become exponentially tiny, so preserving the deterministic path seems to require ϕ exponentially large. That does not yield a superpolynomial lower bound as a function of (N, ϕ) .

4. **Exactify the smoothed FPTAS for approximate equilibria.**

Idea: run $(1 + \varepsilon)$ -improving dynamics with ε below the minimum relative improvement, so that an approximate equilibrium must already be exact.

Obstruction: no polynomial lower bound on the minimum exact improvement is available in unrestricted games; indeed, when $Q(\Gamma)$ is large, the same small-gap issue that obstructs the direct method reappears.

Invoked results

None. Every lemma used in the proof is proved in-line. The web-cited papers were used only for background, orientation, and comparison.

Self-confidence

3

I checked the proof carefully and could not break it, but the compact-network corollary and the precise interpretation of the unrestricted gap are subtle enough that I do not want to claim maximal confidence.

Bibliography

```
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  title     = {The Complexity of Pure Nash Equilibria},
  booktitle = {Proceedings of the 36th Annual ACM Symposium on Theory of Computing},
  pages     = {604--612},
  year      = {2004},
  doi       = {10.1145/1007352.1007445}
}

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  author = {Heiner Ackermann and Heiko Röglin and Berthold Vöcking},
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    title   = {On the Impact of Combinatorial Structure on Congestion Games},
    journal = {Journal of the ACM},
    volume  = {55},
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}

@inproceedings{giannakopoulos2024fptas,
  author = {Yiannis Giannakopoulos},
  title   = {A Smoothed FPTAS for Equilibria in Congestion Games},
  booktitle = {Proceedings of the 25th ACM Conference on Economics and Computation},
  pages    = {401--413},
  year     = {2024},
  doi      = {10.1145/3670865.3673615}
}

@article{chen2019smoothed,
  author = {Xi Chen and Chenghao Guo and Emmanouil-Vasileios Vlatakis-Gkaragkounis and
    Mihalis Yannakakis and Xinzhi Zhang},
  title   = {Smoothed Complexity of Local Max-Cut and Binary Max-CSP},
  journal = {arXiv preprint arXiv:1911.10381},
  year    = {2019},
  eprint  = {1911.10381},
  archivePrefix = {arXiv}
}
\end{verbatim}

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